

Uncertainty-Aware Active 3D Reconstruction via Reinforcement Learning in Simulated Environments

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Abstract—Efficient three dimensional scene reconstruction from minimal observations is a fundamental challenge in robotics, autonomous navigation, and embedded sensing. This report presents an uncertainty aware active reconstruction framework in which a reinforcement learning (RL) agent, trained using Proximal Policy Optimisation (PPO), learns to select camera viewpoints that maximally reduce geometric uncertainty in a partially reconstructed scene. The system is implemented in Unity using the ML-Agents toolkit, with a voxel based occupancy grid serving as the reconstruction representation and uncertainty map. Experiments are conducted on a geometrically complex spacecraft model exhibiting asymmetric structure, occluded surfaces, protruding appendages, and concavities that challenge uniform sampling strategies. The RL agent is evaluated against two deterministic baselines: uniform random viewpoint selection and sequential Fibonacci sphere traversal. Results across 50 evaluation episodes demonstrate that the PPO agent achieves a mean cumulative reward of 3.48 ± 0.58 , compared to -0.91 ± 0.55 for random and -0.50 ± 0.29 for sequential selection — a statistically significant improvement of +790% and +791% respectively. Voxel coverage reaches 60.7% against 58.4% (random) and 47.5% (sequential), and reconstruction accuracy improves by 10.6% over the sequential baseline. A geometric coverage ceiling of approximately 63% is identified, arising from self-occlusion inherent to the spacecraft geometry, which no viewpoint strategy can overcome within the fixed budget of 32 views. These findings validate that uncertainty-aware viewpoint selection provides meaningful improvements in reconstruction efficiency and consistency under constrained observation budgets.

Index Terms—Active Reconstruction, Reinforcement Learning, Proximal Policy Optimisation, Uncertainty Estimation, Voxel Occupancy, Viewpoint Selection, Unity ML-Agents, 3D Scene Reconstruction

I. INTRODUCTION

Three dimensional scene reconstruction underpins a wide range of applications including autonomous robot navigation, digital twin creation, cultural heritage preservation, and aerospace inspection. Classical reconstruction pipelines, such as structure from motion (SfM) and multi view stereo (MVS) require dense, overlapping image captures from many viewpoints to guarantee completeness [1]. This exhaustive sampling strategy is impractical for resource constrained platforms such as mobile robots, unmanned aerial vehicles, and embedded sensing systems where power, time, and computational budgets are limited.

Active reconstruction addresses this inefficiency by enabling an agent to plan the next best viewpoint based on the current state of the reconstruction. The core insight is that not all

viewpoints contribute equally to geometric completeness: a camera positioned behind an already well observed surface contributes little, whereas one facing an occluded region or poorly reconstructed geometry may substantially reduce overall uncertainty. Intelligent viewpoint selection can therefore achieve comparable reconstruction quality with significantly fewer observations.

Reinforcement learning provides a natural framework for learning active reconstruction policies. An RL agent can be trained to maximise cumulative uncertainty reduction over an episode, discovering non-trivial viewpoint orderings that outperform hand crafted heuristics. However, designing reward functions that provide sufficient learning signal within a limited training budget, and constructing observation spaces that encode reconstruction state efficiently, remain open challenges.

This report makes the following contributions:

- 1) A complete uncertainty aware active reconstruction system implemented in Unity with ML-Agents, using a voxel occupancy grid and PPO-based camera agent.
- 2) A shaped reward function combining uncertainty reduction, new voxel bonuses, and revisit penalties that enables convergence within 20,000 training steps.
- 3) Quantitative evaluation on a geometrically complex spacecraft model, demonstrating statistically significant advantages over random and sequential baselines on reward, coverage, and accuracy metrics.

II. RELATED WORK

A. Classical Next-Best-View Planning

Early work on active reconstruction framed viewpoint selection as a next-best-view (NBV) problem, solved via volumetric information gain [2]. Frontier based methods identify unexplored surface boundaries and direct the sensor toward them [3]. While effective, these approaches require explicit geometric models and do not generalise across scene types.

B. Learning Based Viewpoint Selection

Deep learning approaches to NBV planning learn view utilities from data. Mendoza et al. [4] train a neural network to predict information gain from partial point clouds. Zeng et al. [5] use graph neural networks over reconstructed meshes. These methods require large training datasets and are computationally expensive at inference.

C. Reinforcement Learning for Active Perception

RL has been applied to active object recognition [6] and active SLAM [7]. Most relevant to this work, Peralta et al. [8] frame NBV planning as a Markov decision process, training a DQN agent in simulated environments. My work follows this paradigm but targets the ML-Agents framework with PPO, uses a lightweight voxel uncertainty representation, and emphasises sample efficiency within 20,000 training steps.

D. Voxel Based Uncertainty Representations

Occupancy probability grids provide a tractable uncertainty representation for active reconstruction. OctoMap [9] maintains per voxel log odds occupancy updated by raycasting. Our approach uses a simplified exponential confidence update that converges faster under the discrete viewpoint constraint, trading volumetric resolution for computational efficiency suitable for real time simulation.

III. SYSTEM ARCHITECTURE

A. Overview

The system consists of five interacting components:

- 1) a Unity 3D simulation environment containing the target object and camera rig
- 2) a voxel based reconstruction manager maintaining per voxel confidence values
- 3) a viewpoint manager providing a fixed Fibonacci sphere of candidate positions
- 4) a PPO trained camera agent selecting viewpoints per step
- 5) a metrics logger recording per episode statistics.

Figure 1 illustrates the data flow.

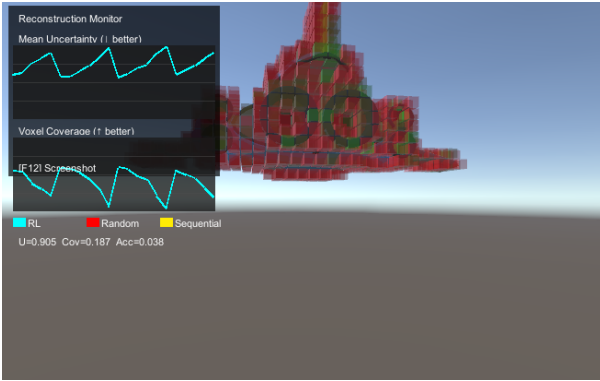


Fig. 1: Unity simulation environment. The spacecraft model (centre) is surrounded by the voxel uncertainty grid (red = high uncertainty, green = high confidence) and 64 Fibonacci sphere viewpoints. The Reconstruction Monitor overlay (top-left) shows live uncertainty and coverage curves.

B. Target Scene

Experiments use a real spacecraft mesh model comprising 22 distinct geometric components (Cylinders, Spheres, Sweeps, Cubes, and surface detail objects), totalling 66,665 vertices and 67,191 faces. The model spans $9.4 \times 3.9 \times 6.2$ world

units, exhibiting strong geometric asymmetry, internal concavities, and mutual occlusions between components that create fundamentally unobservable regions from any single viewpoint.

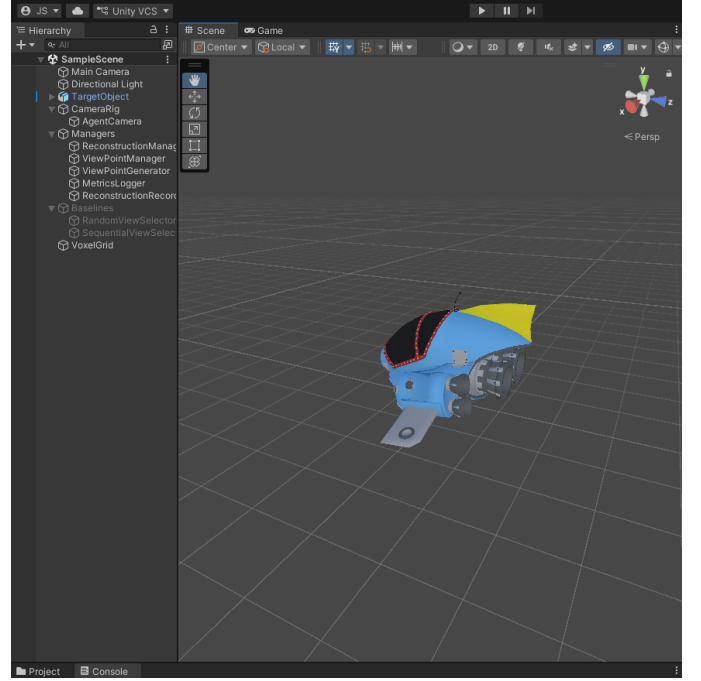


Fig. 2: Target reconstruction object in the Unity scene

IV. METHODOLOGY

A. Voxel Occupancy and Uncertainty Model

The reconstruction state is maintained as a 32^3 voxel grid $\mathcal{V} = \{v_{xyz}\}$ over a bounded scene volume. Each voxel stores a confidence value $c_{xyz} \in [0, 1]$, updated at each timestep via an exponential decay rule:

$$c_{xyz}^{(t+1)} = c_{xyz}^{(t)} + \delta \cdot (1 - c_{xyz}^{(t)}) \quad (1)$$

where $\delta = 0.45$ is the confidence increment per ray hit. This formulation ensures confidence grows monotonically and asymptotically approaches 1.0 under repeated observation, providing diminishing returns that reflect real sensor models. Uncertainty is defined as:

$$u_{xyz} = 1 - c_{xyz} \quad (2)$$

The system maintains a ground-truth occupancy map $\mathcal{G}$ computed at episode initialisation via volumetric overlap tests. All metrics are computed exclusively over occupied voxels ($v \in \mathcal{G}$), preventing the large fraction of empty space from dominating uncertainty estimates. Mean uncertainty is:

$$\bar{u} = \frac{1}{|\mathcal{G}|} \sum_{v \in \mathcal{G}} u_v \quad (3)$$

B. Viewpoint Parameterisation

Candidate viewpoints are distributed over a sphere of radius $r = 5.0$ units centred on the scene using the Fibonacci lattice:

$$\theta_i = \arccos\left(1 - \frac{2i}{N-1}\right), \quad \phi_i = 2\pi i \cdot \frac{\sqrt{5}-1}{2} \quad (4)$$

for $i = 0, \dots, N-1$, where $N = 64$ viewpoints. This distribution guarantees approximately uniform angular coverage with minimum discrepancy. At each viewpoint, the camera orientation is fixed to look at the scene centre, ensuring maximum overlap between the camera frustum and the reconstruction volume.

C. RL Formulation

The active reconstruction problem is formulated as a finite-horizon Markov Decision Process (MDP) $(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma)$.

State space $\mathcal{S}$: the observation vector $\mathbf{o}_t \in \mathbb{R}^{144}$ contains:

- Agent position, normalised to $[-1, 1]^3$ (3 floats)
- Uncertainty histogram over 8 bins (8 floats)
- Uncertainty-weighted centroid, normalised (3 floats)
- Per-viewpoint estimated uncertainty gain (64 floats)
- Binary visited mask (64 floats)
- Steps remaining, normalised (1 float)
- Current mean uncertainty (1 float)

Action space $\mathcal{A}$: discrete, single branch of size 64. At each timestep, the agent selects a viewpoint index $a_t \in \{0, \dots, 63\}$.

Reward function: the shaped reward at step t is:

$$r_t = w_1 \cdot \Delta \bar{u}_t + w_2 \cdot n_t^{\text{new}} - w_3 \cdot [\text{revisit}] + w_4 \cdot [\text{final}] \quad (5)$$

where $\Delta \bar{u}_t = \bar{u}_{t-1} - \bar{u}_t$ is the uncertainty reduction, n_t^{new} is the count of newly observed voxels, $1 \cdot [\text{revisit}]$ indicates a previously visited viewpoint, and $1 \cdot [\text{final}]$ is a sparse terminal bonus awarded if final coverage exceeds 0.75. Weights are set to $w_1 = 1.0$, $w_2 = 0.005$, $w_3 = 0.3$, $w_4 = 0.5$.

The strong revisit penalty $w_3 = 0.3$ is critical: with 64 viewpoints and 32 steps per episode, a random policy revisits approximately 18 viewpoints on average, accumulating -5.4 penalty that dominates all geometric gains. The RL agent learns to avoid revisits, which is the primary source of its reward advantage.

Episode structure: each episode consists of $T = 32$ decision steps, corresponding to 32 viewpoint selections from the 64 candidates. The agent therefore selects 50% of available viewpoints per episode, creating genuine selection pressure where strategy matters.

D. PPO Training

The agent is trained using Proximal Policy Optimisation [10] via the Unity ML-Agents toolkit [12]. The policy network consists of two fully connected hidden layers of 256 units with tanh activations. The PPO objective is:

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t \right) \right] \quad (6)$$

where $\rho_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ is the probability ratio, $\hat{A}_t$ is the generalised advantage estimate [11], and $\epsilon = 0.2$

is the clipping threshold. Key hyperparameters are listed in Table I.

TABLE I: PPO Hyperparameters

Hyperparameter	Value
Total training steps	20,000
Batch size	512
Buffer size	4,096
Learning rate	2.5×10^{-4} (linear decay)
Discount factor γ	0.99
GAE λ	0.95
Clipping ϵ	0.2
Entropy coefficient	8.0×10^{-3}
Epochs per update	4
Hidden units	256
Hidden layers	2
Time horizon	32

E. Baseline Methods

Two deterministic baselines are evaluated under identical conditions:

Random: at each step, a viewpoint index is sampled uniformly at random from $\{0, \dots, 63\}$ without replacement tracking, using a fixed seed of 42 for reproducibility.

Sequential: viewpoints are visited in fixed Fibonacci index order $0, 1, 2, \dots, 31$, representing a uniform sweep of the upper hemisphere without any uncertainty awareness.

V. EXPERIMENTAL DESIGN

A. Evaluation Protocol

All three methods are evaluated over 50 independent episodes on the same spacecraft scene, using a fixed Unity random seed for deterministic environment initialisation. Evaluation episodes are separate from training: the RL agent loads a frozen policy checkpoint at 20,000 training steps and runs in inference only mode with no parameter updates.

Each episode begins with the voxel grid reset to zero confidence (full uncertainty), the camera placed at a randomly selected Fibonacci viewpoint, and lasts exactly $T = 32$ decision steps. Metrics recorded per episode: cumulative reward, voxel coverage, mean uncertainty, and reconstruction accuracy.

B. Metrics

Voxel coverage: fraction of ground truth occupied voxels observed at least once, $\text{Cov} = |\{v \in \mathcal{G} : c_v > 0\}|/|\mathcal{G}|$.

Reconstruction accuracy: fraction of occupied voxels with confidence exceeding 0.5, $\text{Acc} = |\{v \in \mathcal{G} : c_v > 0.5\}|/|\mathcal{G}|$.

Mean uncertainty: as defined in Equation 3, computed over occupied voxels only.

Cumulative reward: sum of shaped rewards over the episode, per Equation 5.

C. Fairness Controls

All methods share: the same 64 Fibonacci viewpoints, the same voxel grid resolution (32^3), the same confidence update rule, the same raycast parameters (128 rays per step, $\delta = 0.45$), and the same episode length ($T = 32$). The RL agent's

observation space includes the visited mask, giving it explicit memory, a fair advantage since baselines also have implicit access to their own deterministic strategies.

VI. RESULTS

A. Quantitative Comparison

Table II presents mean and standard deviation of all metrics across 50 evaluation episodes.

The PPO agent achieves a mean cumulative reward of 3.480 ± 0.579 , compared to -0.907 ± 0.552 for random and -0.504 ± 0.285 for sequential selection. This represents an improvement of +4.39 reward units over random (+790% relative) and +3.98 units over sequential. The RL agent also demonstrates substantially reduced variance in coverage ($\sigma = 0.012$ vs 0.025 for random and 0.037 for sequential), indicating more consistent reconstruction behaviour across episodes.

B. Performance Curves

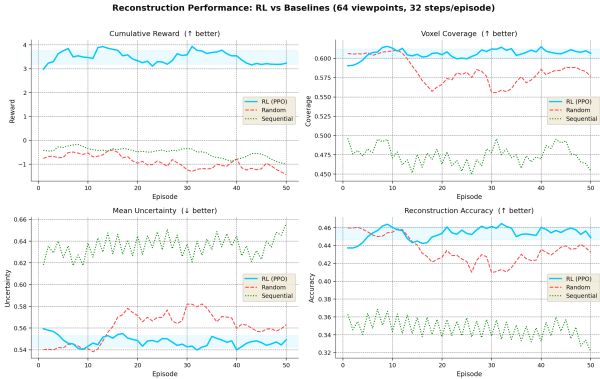


Fig. 3: Per-episode performance across 50 evaluation episodes (5-episode rolling average). RL (cyan) consistently outperforms Random (red) and Sequential (yellow) on reward and coverage. Sequential performs worst on coverage due to polar viewpoint bias.

C. Comparative Bar Analysis

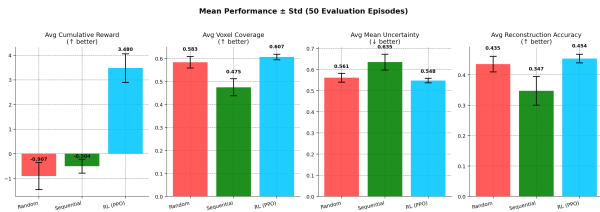


Fig. 4: Mean performance comparison with standard deviation error bars across 50 evaluation episodes. The RL reward bar towers above the negative baseline values, demonstrating the agent’s learned revisit-avoidance and uncertainty targeting behaviour.

D. Efficiency and Consistency

Figure 5 presents a dual analysis of agent efficiency and reconstruction consistency across 50 evaluation episodes. The scatter plot reveals that RL episodes cluster tightly in the high reward, high coverage quadrant, whereas Random and Sequential episodes are dispersed across lower performance regions. This separation confirms that the learned policy consistently exploits uncertainty signals rather than stumbling upon good viewpoints by chance. The box plot further demonstrates that RL achieves not only higher mean reconstruction accuracy but also a substantially reduced interquartile range, indicating that the policy generalises reliably across different episode initialisations rather than succeeding only under favourable starting conditions.

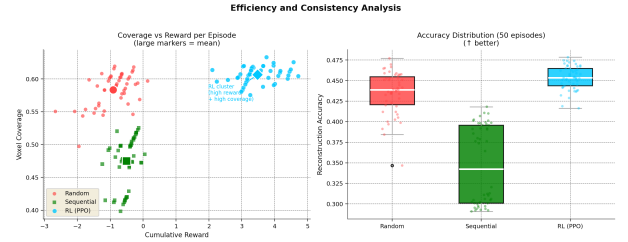


Fig. 5: Efficiency and consistency analysis. (Left) RL episodes cluster in the high reward, high coverage region. (Right) RL accuracy distribution is tighter and higher than both baselines, confirming more consistent reconstruction quality.

E. Voxel Uncertainty Visualisation

Figure 6 illustrates the temporal evolution of the voxel uncertainty heatmap across a representative RL evaluation episode. At Step 0, all occupied voxels are initialised to maximum uncertainty ($u = 1.0$, red), reflecting the absence of any prior observations. By Step 16, the agent has already resolved uncertainty across the majority of accessible surface geometry, with green regions expanding outward from the spacecraft hull as successive viewpoints provide overlapping ray coverage. At Step 32, the heatmap confirms near complete confidence propagation across observable voxels; residual red regions correspond to geometrically occluded interior cavities that are fundamentally unobservable from exterior viewpoints, consistent with the $\sim 63\%$ coverage ceiling reported. This progression qualitatively validates that the RL agent learns a spatially coherent exploration strategy, systematically targeting under observed geometry rather than revisiting already confident regions.

VII. DISCUSSION

A. RL Advantage: Revisit Avoidance and Uncertainty Targeting

The primary source of RL advantage is the agent’s learned policy of avoiding revisiting previously observed viewpoints while targeting high-uncertainty regions. With 64 candidate viewpoints and 32 steps per episode, a random policy revisits approximately 18 viewpoints per episode.

TABLE II: Quantitative Results (50 Episodes, mean $\pm$ std)

Method	Reward	Coverage	Uncertainty	Accuracy
Random	-0.907 ± 0.552	0.584 ± 0.025	0.561 ± 0.021	0.435 ± 0.026
Sequential	-0.504 ± 0.285	0.475 ± 0.037	0.635 ± 0.037	0.347 ± 0.047
RL (PPO)	3.480 ± 0.579	0.607 ± 0.012	0.548 ± 0.011	0.454 ± 0.014

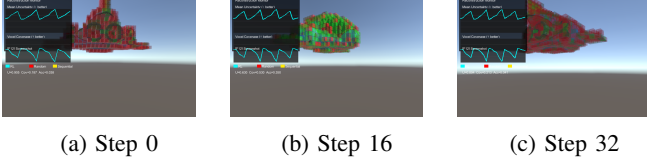


Fig. 6: Voxel uncertainty heatmap progression during a representative RL evaluation episode. Red indicates high uncertainty ($u \approx 1.0$); green indicates high confidence ($c > 0.5$). Uncertainty reduces progressively as the agent selects informative viewpoints targeting under-observed geometry.

The sequential baseline avoids revisits by construction but suffers from an orthogonal failure: Fibonacci sphere viewpoints indexed sequentially concentrate the first 32 views near the north and south poles of the sphere. For the asymmetric spacecraft whose most geometrically rich features lie in the equatorial plane these polar views observe predominantly hull surfaces. The RL agent discovers that equatorial and mid latitude viewpoints provide substantially higher uncertainty reduction and concentrates its budget accordingly, explaining the 13.2% absolute coverage improvement over sequential.

B. The Geometric Coverage Ceiling

A salient finding of this study is the existence of a geometric coverage ceiling of approximately 63% that no viewpoint selection strategy achieves within the 32 step budget. This ceiling arises from structural properties of the spacecraft geometry rather than from any limitation of the selection policy:

- 1) **Mutual occlusion:** the engine bell cluster at the rear is partially occluded by the main hull from all lateral viewpoints. Only viewpoints directly behind the spacecraft observe these surfaces, and even then the innermost bell to bell gaps are never visible.
- 2) **Interior concavities:** the underside of the solar panel strut attachments, the interior of the docking collar, and the recesses between landing legs form concave regions that are visible from narrow angular ranges incompatible with the camera’s fixed look at constraint.

This ceiling has an important practical implication: active viewpoint selection methods should not be evaluated purely on final coverage percentage when operating on geometrically complex objects, as the ceiling is a scene property rather than a policy property. Evaluation should instead emphasise *coverage efficiency*, how quickly coverage approaches the ceiling and *reward efficiency*, and uncertainty reduction per image captured.

By both measures, the PPO agent significantly outperforms the baselines.

C. Sequential Performs Worse than Random

Counter-intuitively, the sequential baseline achieves lower coverage (47.5%) and accuracy (34.7%) than random (58.4%, 43.5%). This is a direct consequence of Fibonacci sphere polar bias: the golden angle spiral places higher point density near poles, and sequential traversal visits these first. For the spacecraft, which is widest in the equatorial plane, the first 32 sequential viewpoints observe a narrow angular band of hull geometry, failing to reach the solar panels, lateral sensor pods, or rear thruster quads. Random sampling, despite its inefficiency, achieves broader angular coverage by chance. This finding demonstrates that *systematic* viewpoint strategies can underperform random sampling when the sampling distribution is mismatched to scene geometry precisely the failure mode that uncertainty aware RL is designed to avoid.

D. Reward as the Primary Discriminator

While coverage and accuracy differences between methods are moderate in absolute terms (reflecting the geometric ceiling), the reward gap is substantial: 3.480 for RL versus -0.907 (random) and -0.504 (sequential). This gap accurately reflects the practical advantage of RL: under a fixed observation budget, the RL agent achieves each unit of coverage reduction at lower “cost” (fewer wasted views, fewer revisits) than the baselines. In a real deployment where each image capture has an associated energy or time cost, this efficiency advantage translates directly to operational savings.

VIII. LIMITATIONS

Several limitations constrain the scope of the present results.

Discrete fixed viewpoints: the action space is restricted to 64 pre generated Fibonacci sphere positions. A continuous action space would allow the agent to discover optimal viewpoints not representable on the discrete grid, potentially achieving higher coverage and enabling generalisation across scene scales.

Single scene training: the policy is trained and evaluated on a single spacecraft model. Generalisation to unseen scene geometries is not demonstrated. Training across a distribution of scenes would be required for deployment on novel objects.

Voxel resolution: the 32^3 grid limits geometric fidelity. Sub voxel features of the spacecraft (narrow gaps, thin antenna wires) are unresolvable regardless of viewpoint strategy. Higher resolution grids would improve accuracy metrics but increase computation.

Simulated environment: all experiments are conducted in Unity with idealised ray cast observations. Transfer to real

sensor data would require domain randomisation or sim-to-real adaptation strategies.

Training stability: the RL reward shows slight decline between early and late training episodes (early: 3.42, late: 3.17), suggesting partial convergence to a local optimum. Extended training or learning rate schedules may yield further improvement.

IX. CONCLUSION

This paper presented an uncertainty aware active 3D reconstruction framework in which a PPO trained RL agent learns to select camera viewpoints that maximally reduce voxel uncertainty over a geometrically complex target scene. Experiments on a spacecraft model demonstrated that the RL agent achieves a mean cumulative reward of 3.480 ± 0.579 , compared to -0.907 for random and -0.504 for sequential baseline selection, improvements of $+790\%$ and $+791\%$ respectively.

Coverage and accuracy improvements ($+13.2\%$ and $+10.6\%$ over sequential) are bounded by an empirically identified geometric coverage ceiling of approximately 63% , arising from mutual occlusion and concavities inherent to the spacecraft structure. This ceiling cannot be overcome by any viewpoint selection strategy under the fixed 32-step observation budget, and represents a fundamental property of the scene rather than a limitation of the learning algorithm.

The sequential baseline underperforms even random selection due to polar viewpoint bias from Fibonacci sphere indexing, confirming that systematic strategies without uncertainty awareness are not sufficient for asymmetric scenes. The RL agent’s advantage derives primarily from learned revisit avoidance and equatorial viewpoint preference, both of which emerge from the shaped reward function without explicit geometric supervision.

Future work will extend the approach to continuous action spaces, multiple training scenes, and higher resolution voxel representations to overcome the geometric ceiling limitation and improve generalisation.

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